NIHAL BAMBALE

Software Engineer | Multimedia & Embedded Application Developer
+91-9420020106 | nihalbambale2000@gmail.com | LinkedIn | GitHub | Hyderabad, India

SUMMARY

Software Engineer with 2+ years of experience in embedded multimedia application development on custom SoC and Linux platforms. Experienced in real-time video and audio streaming, V4L2 based capture pipelines, zero-copy memory design, and Qt/QML application development, with a strong focus on low latency, performance optimization, and end-to-end application delivery.

PROFESSIONAL SUMMARY

- Experienced in application-layer software development for real-time multimedia streaming on custom SoC and Embedded Linux platforms.
- Developed end-to-end streaming pipelines using Live555, RTP/UDP, TurboJPEG compression, and Opus audio encoding.
- Implemented zero-copy memory architectures with V4L2 subsystems using standard and custom IOCTLs to minimize latency during media processing.
- Designed and developed real-time client applications integrating a C++ backend with a Qt/QML frontend for media playback and control.
- Hands-on experience with ALSA based audio capture and playback integration in streaming applications.
- Proficient in Embedded Application Programming, Linux user-space development, IPC, multithreading, and memory management.
- Experienced in debugging and performance profiling using GDB, Valgrind, and Linux debugging utilities.
- Collaborated with driver, platform, validation, and hardware teams for system bring-up, integration, and end-to-end validation.
- Contributed to unit testing, integration testing, code reviews, and Agile software development.
- Proficient in C, C++, Java, Python, SQL, Data Structures, and Object-Oriented Design.

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, SQL (MySQL/Oracle).

Multimedia: Live555, RTP/UDP, RTSP, V4L2, TurboJPEG, Opus, ALSA.

UI Frameworks: Qt, QML, Qt Widgets, Signals & Slots.

Embedded/Linux: Custom SoC platforms, Embedded Application Programming, Linux User-Space Development.

Web/Backend: Spring Boot, HTML/CSS, JavaScript.

Tools: GCC, GDB, Valgrind, Make/CMake, Wireshark, VLC, FFmpeg, strace.

Operating Systems: Linux (Embedded & Application Development), Windows.

Version Control: GitHub, GitLab.

Core Concepts: Data Structures, OOPS, Multithreading, IPC, Memory Management, Socket Programming.

PROFESSIONAL EXPERIENCE

Software Engineer 1 | Multimedia Application Developer
Mirafra Technologies Private Limited | September 2024 – Present

- Engineer application-layer software for robust real-time multimedia streaming using Live555, RTP/UDP protocols, TurboJPEG, and Opus compression.
- Implement zero-copy memory architectures interfacing with V4L2 subsystems via standard and custom IOCTLs to drastically minimize system latency during media processing.
- Develop real-time client applications integrating a C++ backend with a modern Qt/QML frontend that interfaces seamlessly with the streaming architecture.
- Integrate ALSA based audio capture and playback paths for synchronized audio-video streaming.
- Perform root-cause analysis, latency profiling, and performance optimization of media pipelines using GDB, Valgrind, and Wireshark.
- Validate streaming applications on custom SoC platforms and standard Linux environments for stability and throughput.

- Collaborate with driver, platform, hardware, and validation teams for bring-up, integration, and system-level testing.
- Contribute to software quality through unit testing, integration testing, code reviews, and coding standards.
- Participate in Agile/Scrum ceremonies, including sprint planning, daily stand-ups, sprint reviews, and retrospectives.

PROJECTS

1) Real-Time Multimedia Streaming Application

Description

Developed an end-to-end low-latency audio-video streaming application on a custom SoC platform, delivering compressed media over the network to standard RTSP/RTP clients.

Key Responsibilities

- Built the streaming server using the Custom RTP frameworks/Live555 framework over RTP/UDP transport.
- Implemented TurboJPEG based video compression and Opus based audio encoding.
- Handled session management, packetization, and streaming synchronization.
- Performed debugging, latency measurement, and throughput optimization.

Technologies: C++, Embedded Linux, Live555, RTP/UDP, RTSP, TurboJPEG, Opus, Sockets, GDB

2) Zero-Copy V4L2 Video Capture Pipeline

Description

Designed a zero-copy video capture path interfacing with the V4L2 subsystem to eliminate redundant buffer copies and reduce end-to-end media processing latency.

Key Responsibilities

- Implemented V4L2 buffer management using memory-mapped and DMA buffer sharing techniques.
- Interfaced with the capture device using standard and custom IOCTLs.
- Optimized the buffer lifecycle to remove memory copies between capture and encode stages.
- Validated frame rate, latency, and stability across long-duration capture runs.

Technologies: C, C++, Embedded Linux, V4L2, IOCTL, Zero-Copy Buffers, Multithreading

3) Qt/QML Real-Time Streaming Client

Description

Developed a cross-platform desktop client application to receive, decode, and render live media streams with an interactive control interface.

Key Responsibilities

- Implemented a C++ backend for stream reception, decoding, and frame delivery.
- Built a Qt/QML frontend for live video rendering and stream control.
- Integrated backend and UI layers using signals, slots, and threaded rendering.
- Performed debugging, UI responsiveness tuning, and integration testing.

Technologies: C++, Qt, QML, Multithreading, RTP/UDP, ALSA, Embedded Linux

EDUCATION

Master of Technology (M.Tech.) – Artificial Intelligence and Data Science
Indian Institute of Technology (IIT) Patna
2026 – Present

Bachelor of Engineering (B.E.) – Computer Science and Engineering
Rajiv Gandhi College of Engineering and Research
2019 – 2023

CERTIFICATION

Post Graduate Diploma in Advanced Computing (PG-DAC) – 62.00%
CDAC Pune
April 2024 – August 2024

Achievements: HackerRank – C++ (5 Star), Java (4 Star), SQL (4 Star).